



Check the language before you trust the scores

English-trained NLP tools on non-English text — worked example: transforEmotion and Finnish
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Many R packages for sentiment and emotion analysis are built around English language models. They usually accept text in any language and return plausible-looking scores, but the scores may say nothing about the text. This note gives a short checklist for users of social statistics and a worked example: running the R package [transforEmotion](#) on Finnish text.

In short: With its default settings, transforEmotion gave results for Finnish text that were no better than chance. It raised no warning, and the output looked as normal as it does for English text. When we switched to a multilingual model, Finnish results came close to the English baseline.

Checklist: before using an NLP tool on non-English text

1. **Find out which language the default model was trained on.** Read the model card on Hugging Face, not just the package description. "Transformer model" does not mean "multilingual".
2. **Check whether the model can be changed.** If the package accepts any Hugging Face model ID, you can switch to a multilingual one (for zero-shot classification e.g. `MoritzLaurer/mDeBERTa-v3-base-mnli-xnli`).
3. **Look for hidden English.** Prompts, hypothesis templates, label definitions, stop word lists, lexicons and word embeddings are often English-only and fixed in the code, even when the main model can be changed.
4. **Test on a labelled sample in your own language.** About 100–200 examples is enough to tell a working setup from a broken one. Compare the result with the chance level (1 / number of classes).
5. **Look at the distribution of the predictions, not just accuracy.** A model that does not understand the language often puts almost everything in one class. Its average accuracy can still look reasonable.
6. **Report uncertainty.** With 200 test sentences, the 95 % confidence interval for accuracy is about ± 0.06 . Small differences between setups are noise.
7. **Record versions.** Package, model, Python library and R versions all affect the results.

Worked example: transforEmotion and Finnish

What the package does with language

We read the source code of transforEmotion 0.1.7 (GitHub version, January 2026):

Function	Components	Language
<code>transformer_scores()</code>	Zero-shot NLI models; default <code>cross-encoder/nli-distilroberta-base</code>	Defaults English-only; any Hugging Face model can be given
<code>transformer_scores()</code>	Hypothesis template	Fixed to the Hugging Face default <i>"This example is {}."</i> ; the user cannot change it
<code>vad_scores()</code>	NLI models + built-in label definitions	English definitions; custom labels possible
<code>image_scores()</code> , <code>video_scores()</code>	CLIP models	English text encoder
<code>rag()</code> , <code>rag_sentemo()</code>	Small LLMs + retrieval embedding <code>BAAI/bge-small-en-v1.5</code>	Retrieval embedding English-only
<code>sentence_similarity()</code>	<code>all-MiniLM-L6-v2</code>	English-only
<code>nlp_scores()</code> , <code>emoxicon_scores()</code> , stop words	GloVe, TASA, ukWaC and other English semantic spaces and lexicons	English-only

The package documentation does not discuss other languages.

Experiment

Data. [XED](#) (Öhman et al. 2020), movie subtitles annotated with Plutchik's eight basic emotions. We sampled 25 single-label sentences of at least four words per emotion and language: 200 Finnish and 200 English sentences (seed 2026). Chance level is 0.125.

Setups. Zero-shot classification, one predicted emotion per sentence:

	Text	Model	Labels and template
A	English	default (DistilRoBERTa)	English labels (baseline)
B	Finnish	default	English labels (transforEmotion as is)
C	Finnish	mDeBERTa-v3-base-mnli-xnli	English labels
D	Finnish	mDeBERTa	Finnish labels, English template
E	Finnish	mDeBERTa	Finnish labels + Finnish template <i>"Tämä teksti ilmaisee {}."</i>

A–D were run with `transforEmotion::transformer_scores()`. E calls the same Hugging Face pipeline directly via `reticulate`, because the package does not let the user change the template. For E the labels are in the partitive case (*vihaa, iloa, surua...*) so that the template is grammatical. This is a Finnish-specific detail that English-based templates never have to deal with.

Results

Setup	Accuracy	Macro-F1
A English, default model	0.305	0.296
B Finnish, default model	0.130	0.037
C Finnish, mDeBERTa, English labels	0.290	0.272
D Finnish, mDeBERTa, Finnish labels	0.265	0.255
E Finnish, mDeBERTa, Finnish labels + template	0.290	0.268

Number of predictions per emotion (out of 200):

	A	B	C	D	E
anger	3	0	23	6	2
anticipation	90	197	41	22	26
disgust	16	0	2	18	15
fear	6	1	8	16	22
joy	16	0	11	26	11
sadness	12	2	16	24	6
surprise	37	0	64	51	96
trust	20	0	35	37	22

What we learned

- **The default setup fails silently on Finnish.** Setup B classified 197 of 200 Finnish sentences as *anticipation*. Its accuracy (0.130) equals chance. Every sentence still received a full set of scores. Only the distribution of the predictions reveals the problem.
- **A multilingual model largely fixes it.** With mDeBERTa, Finnish accuracy (0.290) was close to the English baseline (0.305). Finnish is not among the 15 XNLI training languages of mDeBERTa, so this result comes from cross-lingual transfer. In transforEmotion the fix is one argument: `transformer = "MoritzLaurer/mDeBERTa-v3-base-mnli-xnli"`.
- **Translating the labels and template did not help.** C, D and E differ by less than the uncertainty (± 0.06). English labels worked as well as Finnish ones for Finnish text.
- **Every setup favoured some class.** The default model favoured *anticipation* even in English, and mDeBERTa favoured *surprise*. With the Finnish template, half of all predictions were *surprise*. A likely cause is that many subtitle lines are questions. Calibrating label scores is worth testing before translating anything.

Limitations

- The absolute level is low in every setup. Short subtitle lines without context are hard to classify, even for human annotators.
- The Finnish and English sentences are different sentences, so A is a rough baseline, not a paired comparison.
- One sample of 200 sentences per language, one domain (film subtitles).
- A machine-translation setup (Finnish $\rightarrow$ English with Helsinki-NLP/opus-mt-fi-en before classification) was planned but did not run in our environment.

Environment and practical pitfalls

Versions used: R 4.6.1, reticulate 1.47.0, transforEmotion 0.1.7, Python 3.14, transformers 5.18.0, torch 2.14.1 (CPU), Windows.

Setting up Python was the hardest part. Notes for others on managed Windows machines:

- transforEmotion builds its Python environment with `uv` through `reticulate::py_require()`. If `uv` is missing or a package fails to install, the error is hidden. You then get `No module named 'transformers'`, even though the setup message says the dependencies were installed.
- If another Python is already selected (`py_config()` shows "forced by use_python()", e.g. through RStudio's Global Options > Python), `py_require()` is ignored.

- University-managed machines may block the unsigned Python builds that uv downloads ("DLL load failed ... blocked by application control policy").

What worked: a named virtual environment built on the officially installed (signed) Python and selected explicitly before the package is loaded:

```
uv venv "$env:USERPROFILE\te-koe314" --python  
"$env:LOCALAPPDATA\Programs\Python\Python314\python.exe"  
uv pip install --python "$env:USERPROFILE\te-koe314\Scripts\python.exe" torch transformers  
sentencepiece protobuf sacremoses
```

```
# in a fresh R session, before library(transforEmotion)  
reticulate::use_virtualenv(file.path(Sys.getenv("USERPROFILE"), "te-koe314"),  
                           required = TRUE)  
library(transforEmotion)
```

Reproduce

- R script: [transforEmotion_kielikoe.R](#) (comments in Finnish)
- Sample: [xed_otos.csv](#) (from XED, CC BY 4.0)

References

- Öhman, E., Pàmies, M., Kajava, K., & Tiedemann, J. (2020). XED: A Multilingual Dataset for Sentiment Analysis and Emotion Detection. *Proceedings of COLING 2020*. [Data](#)
- Tomašević, A., Golino, H., & Christensen, A. P. (2026). transforEmotion: An open-source R package for emotion analysis using transformer-based generative AI models. *Computational Communication Research*. [doi:10.5117/CCR2026.2.2.TOMA](#)
- Laurer, M. mDeBERTa-v3-base-mnli-xnli. [Model card](#)

Experiment run in October 2026. Results depend on model and library versions.